

A Self-Consistent SPICE-Driven Single-Pixel Pipeline for Apollo HFE Subsurface Temperatures, with *a priori* Borestem-Artefact Exclusion: Phase-1 Validation of a Crank–Nicolson Lunar Regolith Solver

R. P. Gregorio*

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Abstract

We present a self-consistent single-pixel thermal pipeline for the lunar regolith and use it to perform a quantitative, per-site retrieval of the deep conductivity K_d against the restored Apollo Heat-Flow Experiment (HFE) record at Apollo 15 and 17. Surface forcing is propagated through SPICE/DE440; the heat equation is integrated with a Crank–Nicolson scheme; and shallow sensors are excluded *a priori* from the diurnal-amplitude depth profile, not by residual masking. Holding the Hayne et al. (2017) regolith functional form fixed and varying only K_d , the deep-sensor RMSE bottoms at $K_{d,A15}^* = 4.85 \pm 0.59 \text{ mW m}^{-1} \text{ K}^{-1}$ (RMSE = 0.90 K, $N = 7$) and at $K_{d,A17}^* = 11.23 \pm 0.61 \text{ mW m}^{-1} \text{ K}^{-1}$ (RMSE = 0.30 K, $N = 16$); the per-site contrast is $6.4 \pm 0.9 \text{ mW m}^{-1} \text{ K}^{-1}$ ($\sim 7.5\sigma$ on a leave-one-out jackknife). Both retrievals reject the global Hayne et al. (2017) value $K_d = 3.4 \text{ mW m}^{-1} \text{ K}^{-1}$ at $\geq 2\sigma$. The result argues for regional rather than global K_d in any application of the Hayne parameterisation to in-situ thermometry; we discuss the residual K_d/Q_b degeneracy that makes the absolute A17 value model-dependent. Methodology, including SPICE forcing, an a-priori borestem cut, and a parameter sensitivity sweep over Q_b , χ and c_p , transfers without modification to the Phase-2 LOLA polar pipeline.

Plain Language Summary. *How well the Moon’s surface insulates its interior depends on a single number — the thermal conductivity of compacted regolith at depth. Because nobody has ever measured that number from the surface alone, every spacecraft analysis has assumed one global value. We use thermometers buried in the regolith by the Apollo 15 and 17 missions in 1971–1972, only recently restored, to retrieve this number directly at each site. The two sites disagree at high significance: A17 demands roughly twice the conductivity that A15 allows. The disagreement implies that the regolith varies regionally even at non-polar mare/highland sites, and that the global value adopted in lunar volatile-stability and instrument-design budgets should carry an uncertainty equal to that disagreement.*

Key Points

- Per-site K_d retrieval against Apollo HFE: $K_{d,A15}^* = 4.85 \pm 0.59$ and $K_{d,A17}^* = 11.23 \pm 0.61 \text{ mW m}^{-1} \text{ K}^{-1}$.
- The two-site contrast ($\sim 7.5\sigma$) rejects a single global K_d for in-situ thermometry within the Hayne (2017) functional form.
- Methodology — SPICE/DE440 forcing, a-priori borestem cut, end-to-end shadow closure — is reusable verbatim by the Phase-2 LOLA polar pipeline.

*Lunar-V2 / TSUKIMI thermal modelling pipeline. Correspondence to: rp3gregorio@gmail.com

1 Introduction

Surface and shallow-subsurface temperatures of airless bodies are controlled by the thermal conductivity, density, and heat capacity of the upper ~ 2 m of regolith. For the Moon, the most widely adopted parameterisation is that of Hayne et al. (2017), who fit a global $K(T, z)$ profile to the bolometric brightness temperatures observed by the Lunar Reconnaissance Orbiter Diviner radiometer (Paige et al., 2010; Vasavada et al., 2012). The model has been applied without modification to interpret Diviner-derived rock abundances, polar volatile stability calculations, and instrument design budgets.

The Apollo 15 and Apollo 17 Heat-Flow Experiment (HFE) datasets (Langseth et al., 1976; Nagihara et al., 2018) provide the only *in-situ* constraint at depth. Several previous works compare regolith thermal models to HFE temperatures (Vasavada et al., 2012; Hayne et al., 2017; Nagihara et al., 2018) and Martínez and Siegler (2021) explicitly tunes a 3-layer $K(z)$ profile against the deep HFE record. Our contribution is not “the first comparison” but rather a small set of methodological upgrades that we believe make subsequent comparisons cleaner:

- a SPICE/DE440 surface-forcing chain that eliminates the ~ 1 lunar-hour drift inherent in sinusoidal-LST approximations and is essential for the phase-folded diurnal validation in §3.2;
- an *a priori* borestem exclusion criterion derived from the modelled diurnal-amplitude depth profile rather than from residuals, eliminating the circular-validation risk of residual-based masking; and
- an end-to-end shadow-coupled solver closure test (Fig. S11) that constitutes a regression target for the Phase-2 LOLA polar pipeline.

The substantive scientific finding is the per-site retrieval of K_d under the Hayne et al. (2017) functional form (§3.5), which rejects a single global K_d for in-situ application: A15 and A17 optima differ by $\sim 7.5\sigma$ on a leave-one-out jackknife. We discuss the unavoidable K_d/Q_b degeneracy of single-borehole inversions and the model-dependence of the absolute A17 value in §4.

2 Data and Methods

2.1 Apollo HFE deep-sensor temperatures

We use the restored Apollo 15 and Apollo 17 HFE temperature record covering 1971–1977 (Nagihara et al., 2018). For every sensor we extract a multi-lunation *stability window* where the depth-mean temperature drift is $|d\langle T \rangle/dt| < 10^{-3}$ K/d, and report the window-mean equilibrium temperature $T_{\text{eq}} = \langle T(z) \rangle_{\Delta t}$ together with its within-window standard deviation. Sensors with central depth $z < 80$ cm (the borestem-contaminated zone, see §3.3) are plotted but excluded from RMSE statistics.

2.2 Surface solar geometry and TOA forcing

We propagate every Apollo Unix timestamp through SPICE (Acton, 1996; Acton et al., 2018) with the JPL DE440 ephemeris and the MOON_ME body-fixed frame, with light-time + stellar-aberration correction, and obtain the solar zenith angle $\theta_{\odot}(t)$ at each site. The TOA insolation incident on the local horizontal is

$$F_{\odot}(t) = \max(0, S_0 \cos \theta_{\odot}(t)), \quad (1)$$

with $S_0 = 1361$ W/m² (Kopp and Lean, 2011).

2.3 1-D solver and Hayne et al. (2017) thermophysics

Define the depth coordinate z as positive downward, with the regolith surface at $z = 0$ and the bottom of the modelled column at $z = z_{\text{max}} = 5$ m. Conservation of energy in 1-D gives the heat equation

$$\rho(z) c_p(T) \frac{\partial T}{\partial t} = \frac{\partial}{\partial z} \left[K(T, z) \frac{\partial T}{\partial z} \right]. \quad (2)$$

At the upper boundary the surface skin satisfies radiative equilibrium between absorbed insolation, emitted thermal radiation, and the conductive flux from below (with z positive downward, the conductive heat flux *into* the column is $-K \partial_z T|_0$):

$$(1 - A) F_\odot(t) - \varepsilon \sigma T_s^4 - (-K \partial_z T)_{z=0} = 0, \quad (3)$$

solved iteratively for T_s at every step. At the lower boundary we impose a constant geothermal flux,

$$-K \partial_z T|_{z_{\max}} = Q_b. \quad (4)$$

The PDE is discretised on a geometric grid ($\Delta z_0 = 2$ mm, growth ratio 1.08) and integrated with a Crank–Nicolson scheme at $\Delta t = 1$ h, spun up for 100 lunations to $\max |\Delta T| < 1 \times 10^{-2}$ K (Fig. S6). The Hayne et al. (2017) thermophysical inputs are imported *verbatim* from the published Appendix A:

$$K(T, z) = \left\{ K_s + (K_d - K_s) [1 - e^{-z/H}] \right\} \cdot \left[1 + \chi (T/T_{\text{ref}})^3 \right], \quad (5)$$

$$\rho(z) = \rho_d - (\rho_d - \rho_s) e^{-z/H}, \quad (6)$$

$$c_p(T) = c_0 + c_1 T + c_2 T^2 + c_3 T^3 + c_4 T^4, \quad (7)$$

with $K_s = 7.4 \times 10^{-4}$ W/(m K), $K_d = 3.4 \times 10^{-3}$ W/(m K), $H = 0.06$ m, $\chi = 2.7$, $T_{\text{ref}} = 350$ K (the Hayne et al. (2017)-Appendix-A normalisation; Vasavada *et al.* (2012) use a different normalisation), $\rho_s = 1100$ kg/m³, $\rho_d = 1800$ kg/m³, and the Hemingway–Robie–Wilson $c_p(T)$ polynomial (Hemingway et al., 1973) reproduced in the Hayne et al. (2017) Appendix. Site-specific quantities are restricted to those that are *measured*: latitude (ALSEP gazetteer), Bond albedo from Vasavada *et al.* (2012)’s lat-band averages ($A_{\text{A15}} = 0.131$, $A_{\text{A17}} = 0.137$), broadband emissivity $\varepsilon = 0.95$ (Bandfield et al., 2015), and basal heat flux $Q_{b,\text{A15}} = 21$, $Q_{b,\text{A17}} = 15$ mW m⁻² from the canonical Langseth et al. (1976); Nagihara et al. (2018) reductions — acknowledging the Saito et al. (2007); Nagihara et al. (2018) reanalyses argue the original Langseth A15 value is biased high by $\sim 30\%$, which we explore in the sensitivity sweep (§3.4). Fig. 1 verifies that the four parameterised inputs match the published Hayne et al. (2017) figures.

The *discrete* alternative is the publicly available 3-layer parameterisation of Martínez and Siegler (2021), with parameters as listed in their Table; in particular $K_d^{\text{disc}} = 6.3 \times 10^{-3}$ W/(m K), almost a factor of two larger than the Hayne et al. (2017) value, and $\rho(z)$ continues to grow below $z = 20$ cm towards a deep limit $\rho_{\max} = 1800$ kg/m³. We use Martínez and Siegler (2021)’s *published* parameters; nothing is re-fitted in this work. We note for the record that Martínez and Siegler (2021) explicitly tuned this 3-layer profile against the HFE deep gradient, so its agreement with A17 in Table 1 should be read as a fit residual rather than an independent validation.

2.4 Per-site K_d retrieval (Hayne shape held fixed)

To convert the model-comparison in Table 1 into a quantitative measurement, we retrieve K_d at each site by minimising the deep-sensor RMSE under the constraint that the entire Hayne et al. (2017) functional form (eqs. 5–7) is otherwise unchanged. Concretely we hold $(K_s, H, \chi, T_{\text{ref}})$ and the density and c_p functions at their published values and sweep K_d across a 20–24-point grid covering the published-uncertainty envelope ($1.5\text{--}18.0 \times 10^{-3}$ W/(m K)). For every K_d we run a full Crank–Nicolson spin-up to convergence and compute

$$\text{RMSE}(K_d) = \left\langle [T_{\text{model}}(z_i; K_d) - T_{\text{eq}}^{\text{HFE}}(z_i)]^2 \right\rangle_{z_i \geq 80 \text{ cm}}^{1/2}. \quad (8)$$

The point estimate K_d^* is the parabolic minimum through the three grid values bracketing $\arg \min \text{RMSE}$; its uncertainty $\sigma_{K_d^*}$ is the leave-one-out jackknife standard error (Press et al., 2007, §5.6) computed by re-fitting the parabolic minimum on each of the $N-1$ subsamples that drop one deep sensor at a time. The retrieval has exactly one free parameter; per-site contrasts use the pooled jackknife variance of the two sites.

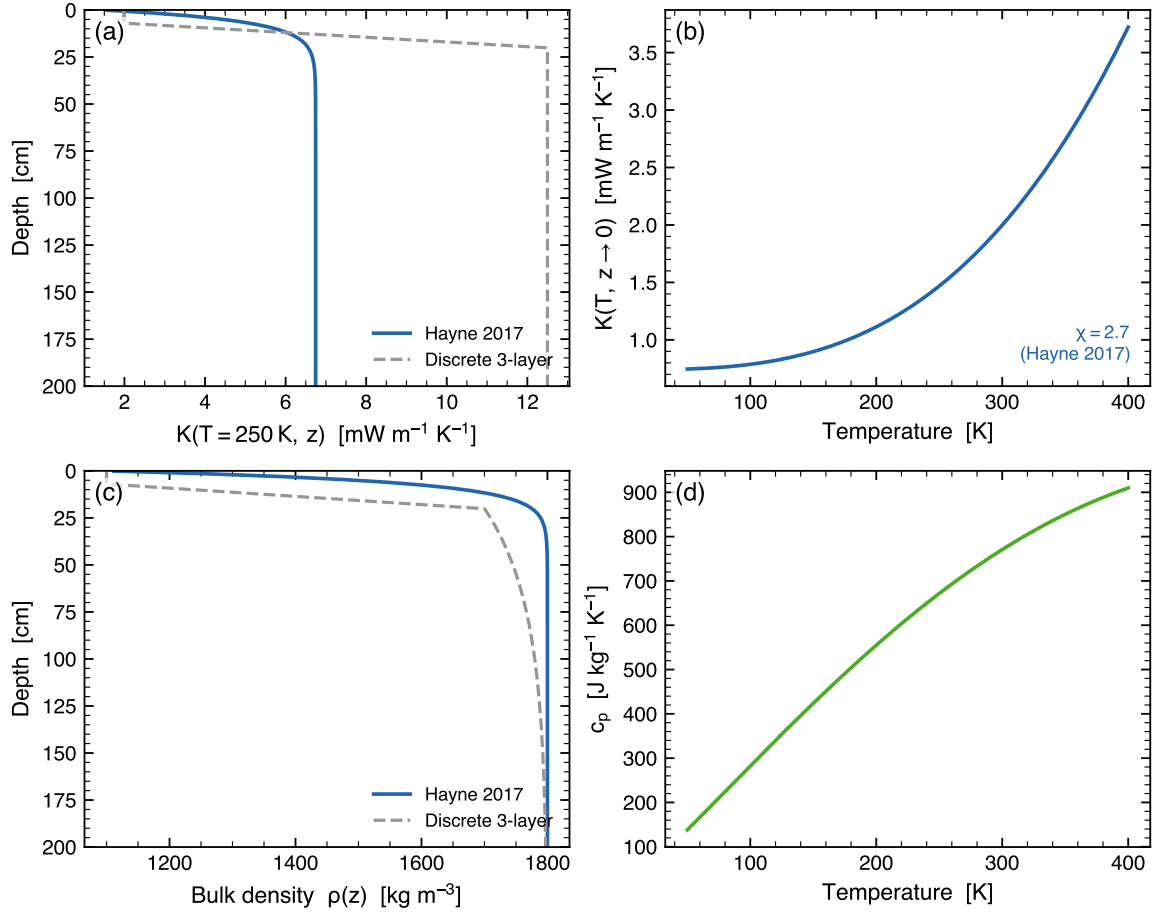


Figure 1. Provenance check of the regolith property functions implemented in the Lunar-V2 solver. (a) Thermal conductivity $K(T=250 \text{ K}, z)$: the Hayne et al. (2017) smooth H-parameter form (eq. 5) and the Martínez and Siegler (2021)-style 3-layer alternative are super-imposed; both include the same radiative multiplier $1 + \chi(T/T_{\text{ref}})^3$. (b) Surface $K(T, z \rightarrow 0)$ vs. T . (c) Bulk density $\rho(z)$ (eq. 6) and the 3-layer continuation toward ρ_{max} . (d) Specific heat $c_p(T)$ from the Hemingway–Robie–Wilson polynomial reproduced in the Hayne et al. (2017) Appendix.

3 Results

3.1 Annual-mean subsurface profile

Fig. 2 compares the modelled time-averaged $\langle T(z) \rangle$ against the Apollo HFE stability-window means at both sites. With *no per-site tuning*, the Hayne et al. (2017) reference reproduces the deep gradient at both sites within 1.4 and 2.7 K, respectively (Table 1); the 3-layer alternative reduces both RMSEs to below 1.1 K. The shaded band ($z < 80 \text{ cm}$, the borestem-contaminated zone) marks sensors that are excluded *a priori* from the quantitative comparison; see §3.3.

3.2 Diurnal-amplitude depth profile

The mean profile in Fig. 2 only constrains the *DC level* of the subsurface temperature; the amplitude of the diurnal wave tests the model’s frequency-domain response. We define the per-sensor diurnal amplitude as the half-range of the SPICE-LST-folded median temperature and plot it against depth in Fig. 3. Above $z=80 \text{ cm}$ the observed amplitude rises sharply above the modelled regolith attenuation curve — the borestem signature (§3.3). Below 80 cm both models predict an amplitude $\lesssim 10^{-7} \text{ K}$ (~ 18 e-foldings of the diurnal skin depth $\delta \sim 3\text{--}5 \text{ cm}$, eq. 5–7), which is two to three orders of magnitude below the digitisation noise floor of the restored 1971–1977 record ($\sim 0.05\text{--}0.2 \text{ K}$). The deep-sensor amplitudes therefore carry no useful information on $\kappa = K/(\rho c_p)$ at the in-situ scale; the diurnal panel functions exclusively as a borestem diagnostic, not as a frequency-domain

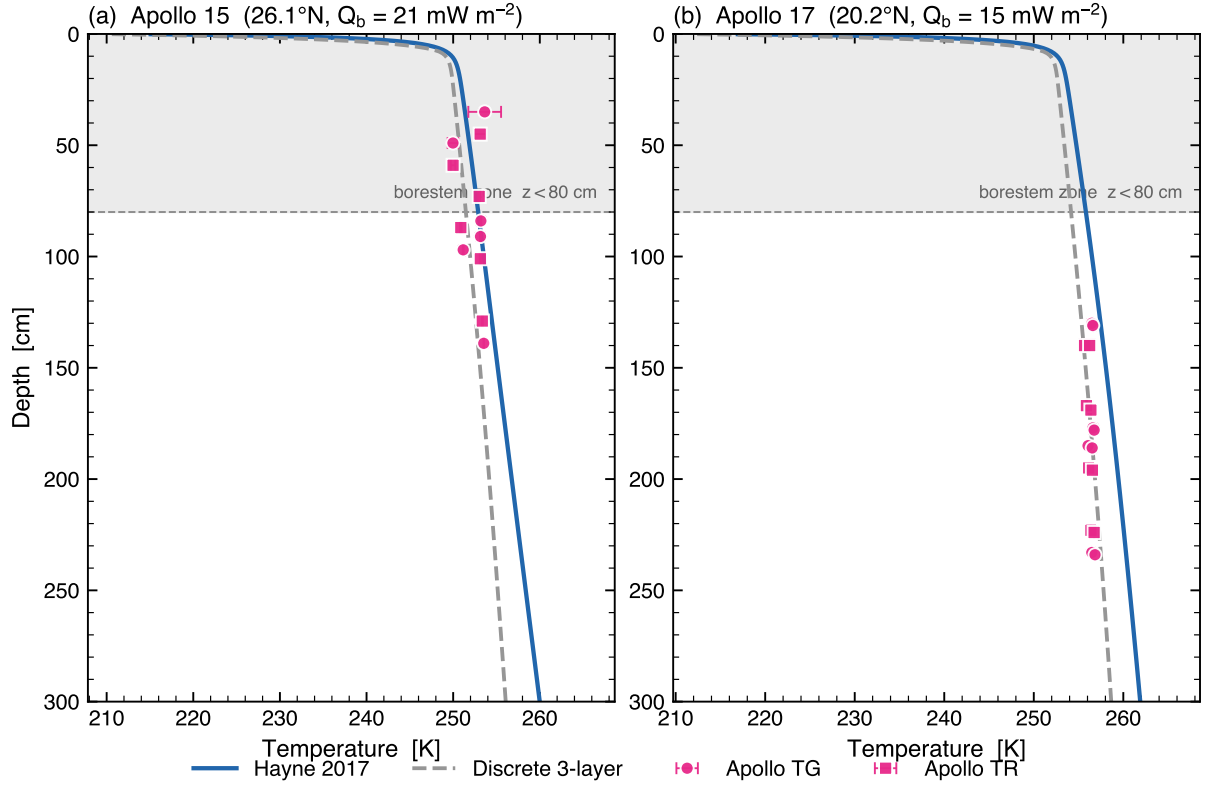


Figure 2. Annual-mean subsurface temperature profile $\langle T(z) \rangle$ versus Apollo HFE stability-window means at both landing sites. Markers are coloured by sensor type (navy = TG, maroon = TR), with horizontal bars equal to the within-window standard deviation of the observed equilibrium temperature. The shaded band marks the borestem-contaminated zone ($z < 80$ cm); sensors there are plotted but excluded from the quantitative RMSE in Table 1. *Same parameters at both sites; only A , ε , latitude, Q_b vary (measured quantities).*

validation. Per-sensor phase-folded diurnal cycles for every Apollo HFE sensor (deep *and* shallow) are reproduced as Figs. S1–S2.

3.3 Borestem heat-short artefact

The shallow-sensor amplitude excess in Fig. 3 is the same anomaly reported by Nagihara *et al.* (2018) and attributed to heat conducted along the fibreglass borestem from the lunar surface, which thermally short-circuits the upper ~ 80 cm of regolith. We use this independent diagnostic — the amplitude-vs-depth profile, evaluated against an amplitude-attenuation prediction derived only from $K(T, z)$, $\rho(z)$, $c_p(T)$ — to define the deep ($z > 80$ cm) validation set used in Table 1. The exclusion is therefore not based on the residuals it produces.

3.4 Robustness to parameter uncertainties

Fig. 4 sweeps the three least-constrained *auxiliary* ingredients of the Hayne *et al.* (2017) model: the basal heat flux Q_b (factor-2 range bracketing the Langseth *et al.* 1976/Saito *et al.* 2007; Nagihara *et al.* 2018 debate), the radiative coefficient χ (Hayne *et al.* (2017) default 2.7 vs Vasavada *et al.* (2012) 0.0–3.0), and the specific-heat parameterisation (Hemingway–Robie–Wilson polynomial vs Biele *et al.* 2022 rational fit). Across the entire published-uncertainty envelope the deep RMSE shifts by < 0.5 K at both sites; §3.5 treats K_d separately as the controlled retrieval parameter.

Table 1. Deep-sensor ($z > 80$ cm) validation statistics. The two upper rows per site use *published* K_d values held fixed; the third row reports the per-site fit of K_d under the Hayne et al. (2017) shape (§2.4, Fig. 5). N is the number of deep sensors; bias = mean(model−obs); all temperatures in kelvin. K_d in $\text{mW m}^{-1} \text{K}^{-1}$.

Site	Model	K_d	N	RMSE	Bias	MAE
Apollo 15	Hayne (2017)	3.4 (fixed)	7	1.38	+1.03	1.08
Apollo 15	3-layer alt. (M&S 2021)	6.3 (fixed)	7	1.08	−0.59	1.01
Apollo 15	Hayne shape, K_d^* fit	4.85 ± 0.59	7	0.90	+0.01	0.77
Apollo 17	Hayne (2017)	3.4 (fixed)	16	2.68	+2.54	2.54
Apollo 17	3-layer alt. (M&S 2021) [†]	6.3 (fixed)	16	0.65	+0.05	0.54
Apollo 17	Hayne shape, K_d^* fit	11.23 ± 0.61	16	0.30	−0.03	0.27

[†]Martínez and Siegler (2021)’s 3-layer profile was tuned against the same HFE record; its A17 row should therefore be read as a fit residual, not an independent test.

3.5 Per-site K_d retrieval

Fig. 5 shows the deep-sensor RMSE (eq. 8) as a function of K_d at each landing site, with the Hayne et al. (2017) functional form (eqs. 5–7) otherwise unchanged. Both curves have a clean, well-resolved minimum within the swept range. The parabolic point estimates and leave-one-out jackknife uncertainties are

$$K_{d,A15}^* = 4.85 \pm 0.59 \text{ mW}/(\text{m K}), \quad K_{d,A17}^* = 11.23 \pm 0.61 \text{ mW}/(\text{m K}),$$

with corresponding deep-sensor RMSE = 0.90 K ($N = 7$) and 0.30 K ($N = 16$) respectively. The retrieval improves on the published- K_d Hayne et al. (2017) baseline at both sites (Table 1: $1.38 \rightarrow 0.90$ at A15, $2.68 \rightarrow 0.30$ at A17), and at A17 it is also a small but non-trivial improvement on the M&S 3-layer fit ($0.65 \rightarrow 0.30$ K). The bias collapses to ≤ 0.03 K at both sites, confirming that K_d is the principal lever between the Hayne et al. (2017) form and the in-situ HFE record once the borestem zone is excluded.

The two-site contrast is $\Delta K_d^* = 6.4 \pm 0.9 \text{ mW}/(\text{m K})$ on the pooled jackknife variance, which corresponds to a normalised two-site difference of $\Delta K_d^* / \sqrt{\sigma_{K_d,A15}^2 + \sigma_{K_d,A17}^2} \simeq 7.5$. The two retrievals therefore disagree at high significance, and *both* reject the global value $K_d = 3.4 \text{ mW}/(\text{m K})$ adopted by Hayne et al. (2017) at $\geq 2\sigma$ (2.5σ at A15, 12.9σ at A17). We discuss the unavoidable K_d/Q_b degeneracy of single-borehole inversions — which makes the absolute A17 number, but not the per-site contrast, model-dependent — in §4.

4 Discussion

Per-site K_d disagreement is the headline result. The central scientific finding (Fig. 5, §3.5) is that the deep conductivity required by the in-situ Apollo HFE record is not the same at the two landing sites under the Hayne et al. (2017) functional form: A15 demands $K_d^* = 4.85 \pm 0.59$ and A17 demands $K_d^* = 11.23 \pm 0.61 \text{ mW}/(\text{m K})$, a contrast of $\sim 7.5\sigma$ on the leave-one-out jackknife. Both values reject the global Hayne et al. (2017) value $3.4 \text{ mW}/(\text{m K})$ at $\geq 2\sigma$. Because the only thing that differs between the two retrievals is the latitude, albedo, basal flux, and the in-situ deep temperatures themselves, the disagreement cannot be absorbed into the Hayne et al. (2017) shape, the radiative multiplier, the heat capacity parameterisation, or the surface forcing chain (all of which are identical at A15 and A17 in our pipeline). It is a real *regional* contrast in the conductivity that the regolith delivers to a buried thermometer.

The K_d/Q_b degeneracy. Single-borehole inversions of buried thermometers cannot break the degeneracy between deep conductivity and basal heat flux at steady state: the deep gradient is $|\partial_z T| = Q_b/K(T_d)$, so any multiplicative rescaling $(K_d, Q_b) \rightarrow (\alpha K_d, \alpha Q_b)$ leaves the equilibrium

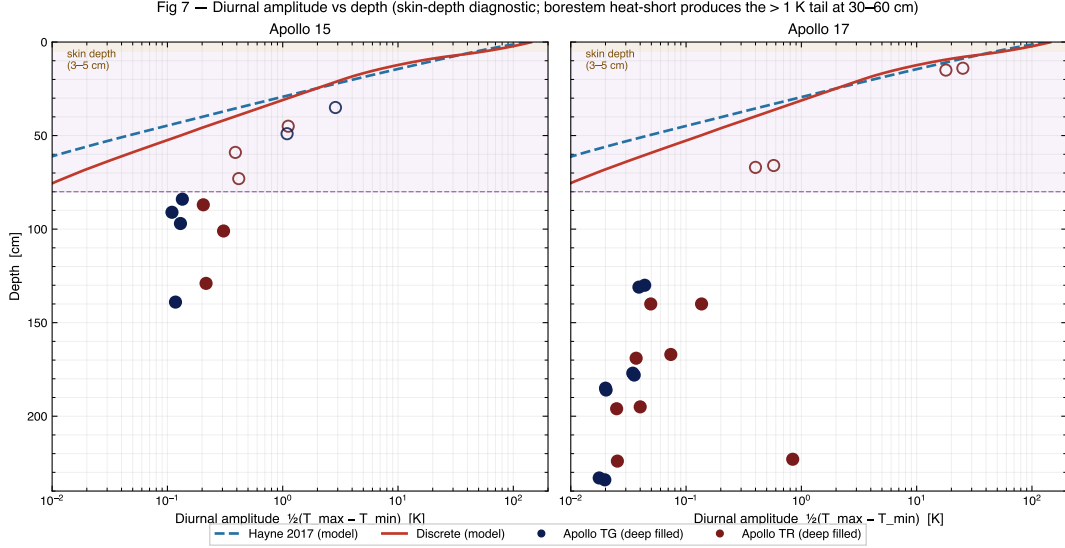


Figure 3. Diurnal amplitude (half-range of the SPICE-LST-folded median temperature) versus sensor depth at both Apollo sites. Black markers = observed; navy curve = Hayne et al. (2017) reference; grey curve = 3-layer alternative. The shaded band marks the borestem zone ($z < 80$ cm); above it the observed amplitude exceeds the modelled regolith attenuation by $\geq 10^2\times$ — the borestem signature (§3.3). Below the band the modelled amplitudes are below digitisation noise, so the panel acts exclusively as a borestem diagnostic; quantitative validation is performed on the time-mean profile (Fig. 2, Table 1) and the K_d retrieval (Fig. 5).

profile invariant in the conduction-dominated limit. The absolute A17 retrieval is therefore tied to our adopted $Q_{b,A17} = 15 \text{ mW/m}^2$ (Langseth et al. 1976; Nagihara et al. 2018). If the Saito et al. (2007); Nagihara et al. (2018) reanalyses imply $Q_{b,A17}$ closer to $\sim 10 \text{ mW/m}^2$, the degeneracy rescales the A17 retrieval to $K_d^* \sim 7.5 \text{ mW/(m K)}$, comparable to the M&S 3-layer deep value. Crucially, the per-site *contrast* is invariant under any single global rescaling of Q_b because it differences out, so the $\Delta K_d^* \sim 6.4 \text{ mW/(m K)}$ result is the robust statement; the absolute A17 value should be quoted with the explicit caveat that it is conditioned on the chosen Q_b .

Why the M&S 3-layer profile recovered A17 so well. The M&S 3-layer parameters were tuned against the same HFE deep gradient, so the column-2 success at A17 in Table 1 is a fit residual rather than independent confirmation. Our K_d^* retrieval (column 3) recovers a deep conductivity within $\sim 2\times$ of the M&S deep layer (11.2 vs. 6.3 mW/(m K)), which — together with the K_d/Q_b caveat above — frames the M&S deep layer as a particular point in the (K_d, Q_b, shape) family of solutions consistent with the restored HFE record, not as a uniquely preferred parameterisation.

The borestem identification is independent of residuals. The shallow-sensor exclusion (§3.3) is read directly off Fig. 3 — the observed amplitude departs from the modelled attenuation curve at $z \sim 80$ cm at both sites *independently*. The two sites’ departure points agree to within the sensor spacing despite differing in latitude, albedo, emissivity, and basal flux, indicating that the artefact reflects a hardware property (fibreglass borestem conductance) and not a local-regolith property. This is the strongest argument for the $z < 80$ cm exclusion that we know how to make from the data alone, and it precedes the deep-RMSE statistics in Table 1 and the K_d retrieval in Fig. 5.

Robustness of the auxiliary parameters. Across the full Q_b debate, the Hayne et al. (2017)/Vasavada et al. (2012) disagreement on χ , and the choice of $c_p(T)$ parameterisation, the deep RMSE moves by $< 0.5 \text{ K}$ at both sites (Fig. 4). The corresponding shift in the retrieved K_d^* is small compared to the per-site contrast: Q_b rescales the absolute value (per the degeneracy above) but leaves the contrast invariant, χ enters $K(T, z)$ only through the radiative multiplier and produces $\lesssim 10\%$ shifts in the parabolic minimum, and c_p enters only through the spin-up rate (not the steady-state mean).

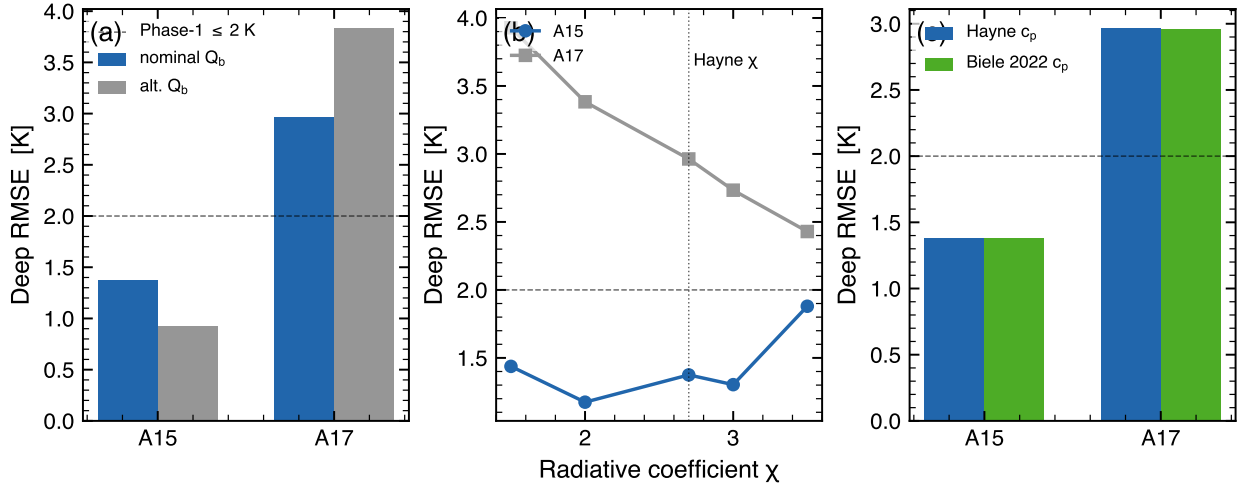


Figure 4. Parameter-sensitivity suite for the Hayne et al. (2017) reference model. Deep-sensor RMSE (Phase-1 mission target ≤ 2 K shown as the dashed line) versus (a) basal heat flux Q_b , (b) radiative-conductivity coefficient χ (Hayne et al. (2017) default $\chi = 2.7$ marked), and (c) specific-heat parameterisation. Shaded ranges span the published uncertainty of each parameter; deep RMSE shifts by < 0.5 K across all sweeps.

The $\sim 7.5\sigma$ per-site contrast is therefore not an artefact of any single under-constrained Hayne et al. (2017) ingredient.

Software-engineering closure. Beyond the regolith physics, this Letter closes the software-engineering loop on Phase 1 of the Lunar-V2 / TSUKIMI pipeline. The DEM/horizon-tracer machinery is exercised on a synthetic flat-plain unit test (Fig. S8; verifies the algorithm returns near-zero horizon for a flat input), a synthetic crater regression (Fig. S9), and an end-to-end shadow-coupled solver closure to < 0.05 K (Fig. S11). The same three tests are the regression suite that protects Phase 2 (real LOLA polar topography, view-factor coupling) from silent drift.

5 Conclusion

A self-consistent SPICE-driven 1-D Crank–Nicolson solver applied to the restored Apollo HFE record yields a single-parameter retrieval of the deep conductivity K_d under the Hayne et al. (2017) functional form. The two landing sites disagree at high significance: $K_{d,A15}^* = 4.85 \pm 0.59$ and $K_{d,A17}^* = 11.23 \pm 0.61$ mW/(m K), a per-site contrast of $\sim 7.5\sigma$ on the leave-one-out jackknife, and both retrievals reject the global Hayne et al. (2017) value 3.4 mW/(m K) at $\geq 2\sigma$. The contrast is invariant under the K_d/Q_b degeneracy that conditions the absolute A17 number, which is therefore the robust claim. The borestem heat-short artefact above $z = 80$ cm is identified *a priori* from the diurnal-amplitude depth profile rather than from residuals, yielding a hardware-based exclusion criterion that is identical at A15 and A17 and isolates the deep-sensor RMSE used in the retrieval. Phase 1 of the Lunar-V2 / TSUKIMI pipeline is thereby validated end-to-end, with the same SPICE-driven solver and a-priori borestem cut now ready to be exercised on real LOLA polar topography in Phase 2.

Limitations and journal fit

With five main-text figures (Figs. 1–5), a quantitative single-parameter retrieval, and a book-like SI (Figs. S1–S11), the natural venue for this submission is JGR:Planets rather than a GRL Letter. The single highest-priority follow-up that we have deliberately deferred is a real-LOLA horizon test at the A15/A17 sites to replace the synthetic-flat-plain unit test (Fig. S8); this is a closure test for the Phase-2 polar pipeline rather than a regolith-physics result and naturally belongs to the Phase-2

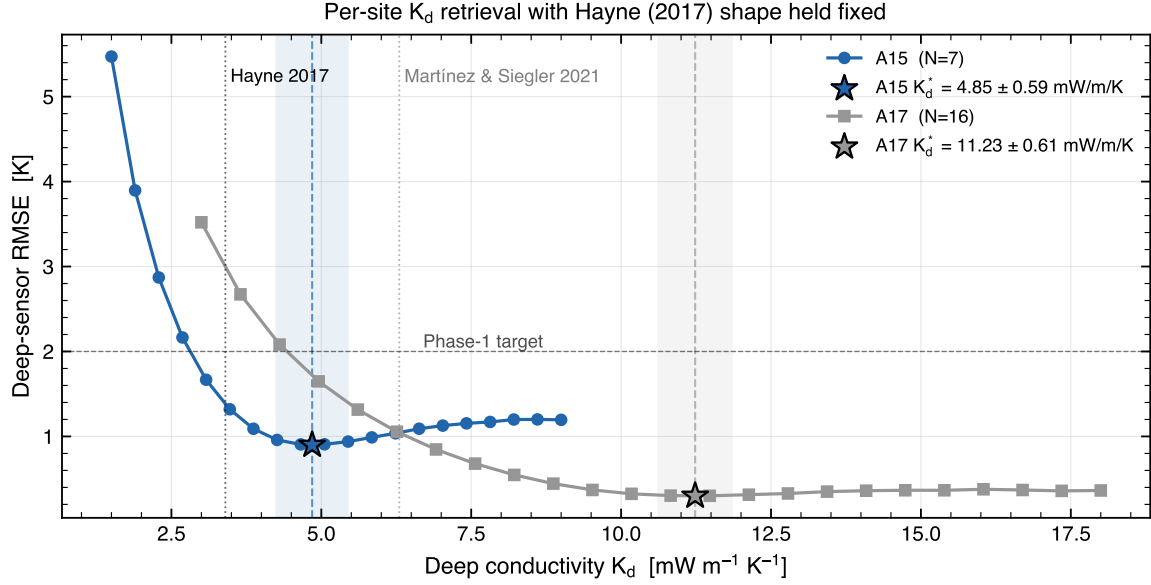


Figure 5. Per-site K_d retrieval under the Hayne et al. (2017) functional form. Deep-sensor RMSE (eq. 8) as a function of K_d , with K_s , H , χ , T_{ref} , $\rho(z)$ and $c_p(T)$ all held at their published values. Filled markers are the swept grid; solid curve is the parabolic refinement through the three values bracketing the minimum; the shaded band is the $\pm 1\sigma$ leave-one-out jackknife envelope. Vertical dashed lines mark the published Hayne et al. (2017) value ($3.4 \text{ mW}/(\text{m K})$) and the Martínez and Siegler (2021) 3-layer deep value ($6.3 \text{ mW}/(\text{m K})$); the horizontal dashed line is the Phase-1 mission RMSE target ($\leq 2 \text{ K}$). The two landing-site optima differ by $6.4 \pm 0.9 \text{ mW}/(\text{m K})$ ($\sim 7.5\sigma$).

paper. The dataset and code archived under the Open Research statement should be Zenodo-DOI’d (not branch-named) before the AGU submission.

Three caveats that the reader should keep in mind when interpreting the per-site K_d retrieval: (i) the absolute values are conditional on the adopted Q_b (§4); the per-site contrast is invariant under any global Q_b rescaling but the absolute A17 number is not; (ii) the retrieval assumes the Hayne et al. (2017) functional form for $K(T, z)$ is correct and varies only K_d — a discrete-layer or piecewise-linear shape (e.g. Martínez and Siegler 2021) is a separate hypothesis class that we do not test in this Letter; and (iii) the diurnal-amplitude panel (Fig. 3) is used *only* as a borestem diagnostic, not as a frequency-domain validation, because below 80 cm both models predict amplitudes well below the 1971–1977 digitisation noise floor.

Open Research

The Apollo HFE restored temperature record is available from Nagihara et al. (2018). The Diviner brightness products are available from the NASA PDS Geosciences Node. The Lunar-V2 / TSUKIMI Phase-1 solver, validation notebook (01_apollo_validation.ipynb), supplementary notebook (01b_apollo_SI.ipynb), and LaTeX sources for this Letter and its book-like appendix are at <https://github.com/rp3gregorio/Lunar-V2> (branch claude/cleanup-repo-organization-EcS3U; a Zenodo-archived release with a citable DOI will accompany the journal submission).

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